

DronePayload Team 4 Teamwork Agreement

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: Drone Payload Team 4

Instructors: Dr. Esperanza Linares Guerrero, Dr Harish Chintakunta

Program/Class: Engineering Pathways Program (EPP)

Members and Roles

1. Peter Yang - Analysis/Validation Lead
2. Sheryl Madlangbayan - Modeling Lead
3. Ezekiel Savage Documentation & Visualization Lead
4. Ryan Nguyen (PM), QA, Analysis assist.

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations

- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: **Discord**
- Secondary channel: none
- Response time: **4hours** on weekdays and **6hours** on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)

Weekdays

Sheryl - (9am-6pm working) but i check my discord regularly

Ezekiel - A few hours response time

Ryan - Ping me whenever besides (12Am-9Am)

Peter - (11am - 1pm, 7pm - 12am)

Weekends

Sheryl - Free (?)

Ezekiel - Free most weekends

Peter - Free

Ryan - Free mostly

- Meeting cadence: **Mondays and thursday 9-10:30**
- Decision records: Refer to Drone "Meeting Notes"

4) Collaboration & Tools

- Version control: Shared GitHub Repository and Ryan Nguyen (PM) organizes GitHub.
- Data management: All datasets, parameters, and assumptions documented in

README.md

File naming & style: **Depends, but descriptive names and Type_arm**

5) Quality Standards (Definition of 'Done')

- Have 2 drone arms that meet requirements
 - (Max Payload = 0.5Kg, Factor of safety ≥ 1.5 , Thrust to Weight Ratio [TWR] $\geq 2:1$)
 - As a bonus, have the drone arms account for commercially available payloads when considering payload weight.
- Have a program that runs TWR and FEA on at least 2 drone arms
 - Have it produce a table with max displacement, max stress, factor of safety, cost of 4 drone arms, Maximum payload weight, and the name of the material the arm has those stats with.
 - Make sure it has a title for each table to distinguish between them
 - Have a visualizer for the FEA results (Use the built in one)
- Have a final report to submit with at least 2 drone arms and a recommendation for the choice.
 - Do the option extension of cost analysis
 - Consider commercially available payloads.
- The program can be run from download by other users.
- Documentation of meetings, drawings, and models
- Fine Tuning of FEA results to have a reasonable amount of error

6) Timeline & Milestones

- Week 1 (July 6 - 11):
 - Start thinking about drone arm designs
 - (everyone will present 1 design by Monday meeting)
 - list out the equations we'll be using,
 - familiarize with matlab, fea, solidworks, github.
- Week 2 (July 12 -18):
 - Pitch drone arm designs
 - Start adding drone arms into MATLAB programs
 - Refinement of Drone arm designs
- Week 3 (July 19 - 25):
 - Start FEA on Solidworks then integrate design to matlab
- Week 4(July 26-Aug 1):
 - Review scoring detail for report
 - Bring documentation of work
 - Finalize drone arm designs and FEA results

- Prepare for a report framework to use
- Week 5 (Aug 2-7)
 - Finish organization of GitHub
 - Finish Final Report

7) Decision-Making & Conflict Resolution

- Default: **Discuss in meeting**
- If blocked >48 hrs: **PM handles dispute or hold a meeting to discuss what to do.**
- Technical disagreements: **Evidence-based comparison before vote**
- Escalation: **If unresolved, consult instructor/mentor with a concise memo.**

8) Workload, Accountability & Attendance

- Task board: **Discussed in meetings and tasks are given in meeting notes. You can organize the tasks given to you in your own way.**
- Accountability: **Discuss during meeting to reorganize tasks**
- Attendance: **Notify others and send over progress of tasks based on previous meetings. Also check meeting notes for task given if you missed it.**

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- ☒ Have 2 drone arms that meet requirements
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 - ☒ As a bonus, have the drone arms account for commercially available payloads when considering payload weight.
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- ☒ The program can be run from download by other users.
- ☒ Documentation of meetings, drawings, and models
- ☒ Fine Tuning of FEA results to have a reasonable amount of error

12) Agreement & Signatures

Name & Signature: **Peter Yang** Date: **7/8/2026**

Name & Signature: **Sheryl Madlangbayan** Date: **7/8/26**

Name & Signature: **Ezekiel Savage** Date: **7/8/2026**

Name & Signature: **Ryan Nguyen** Date: **7/8/26**